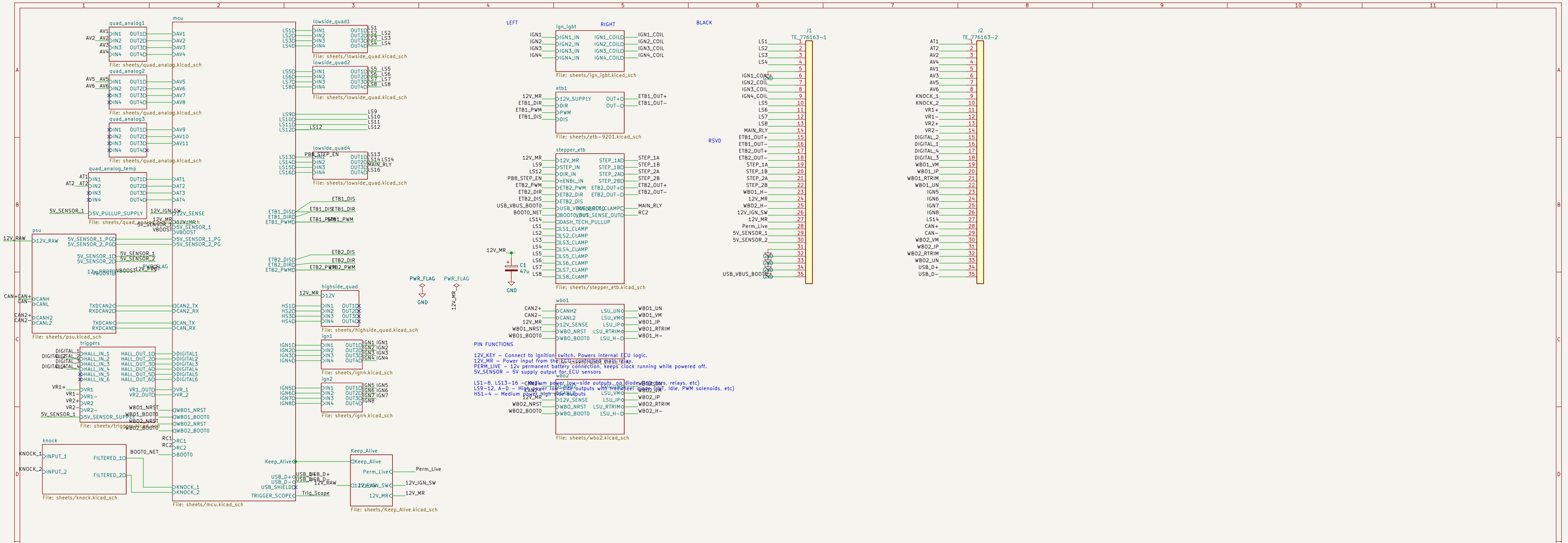
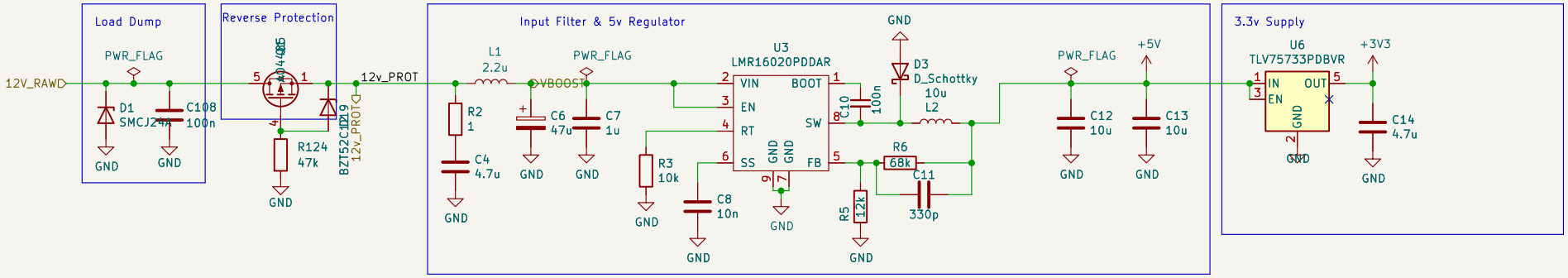
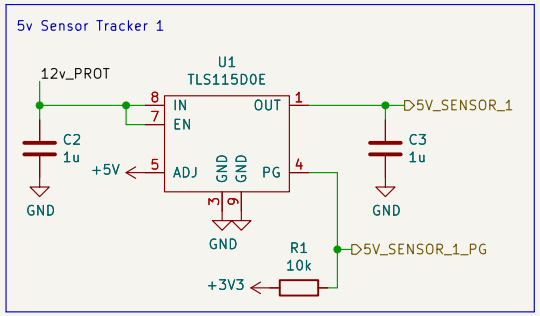


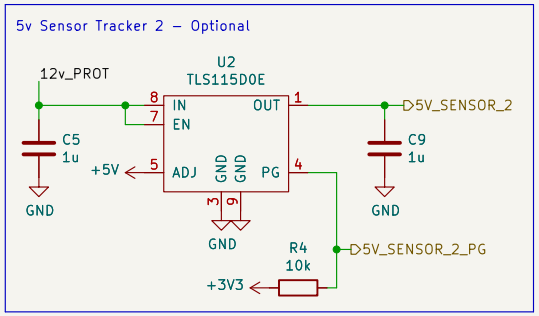
Sheet: /
File: fome-mc.kicad_sch

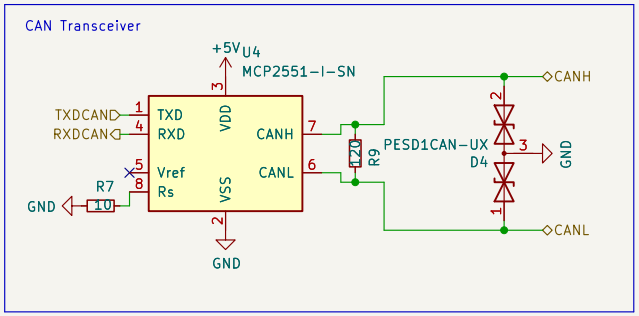
Title: fome-mc Integrated Motorcycle ECU

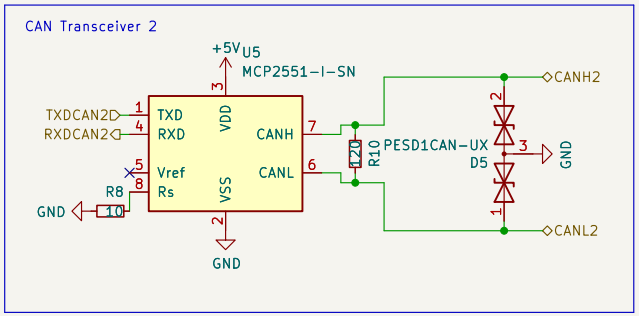
Size: A2	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 1/21









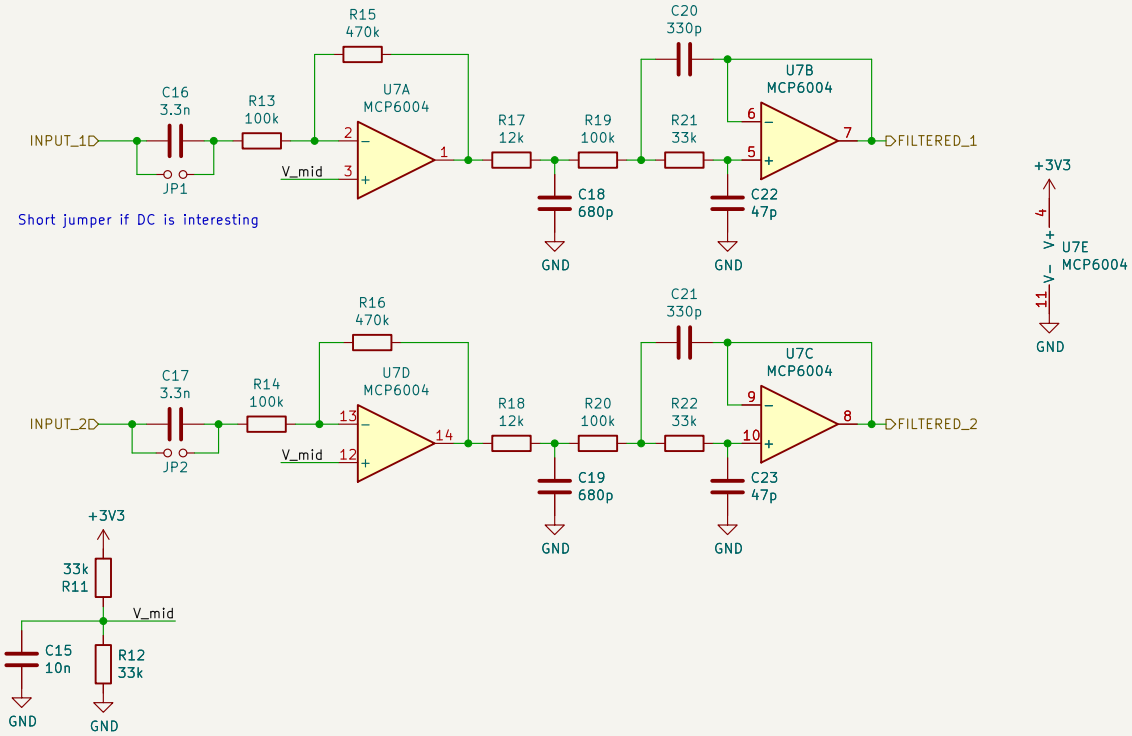


rusefi.com/s/proteus

Sheet: /psu/
File: psu.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 2/21

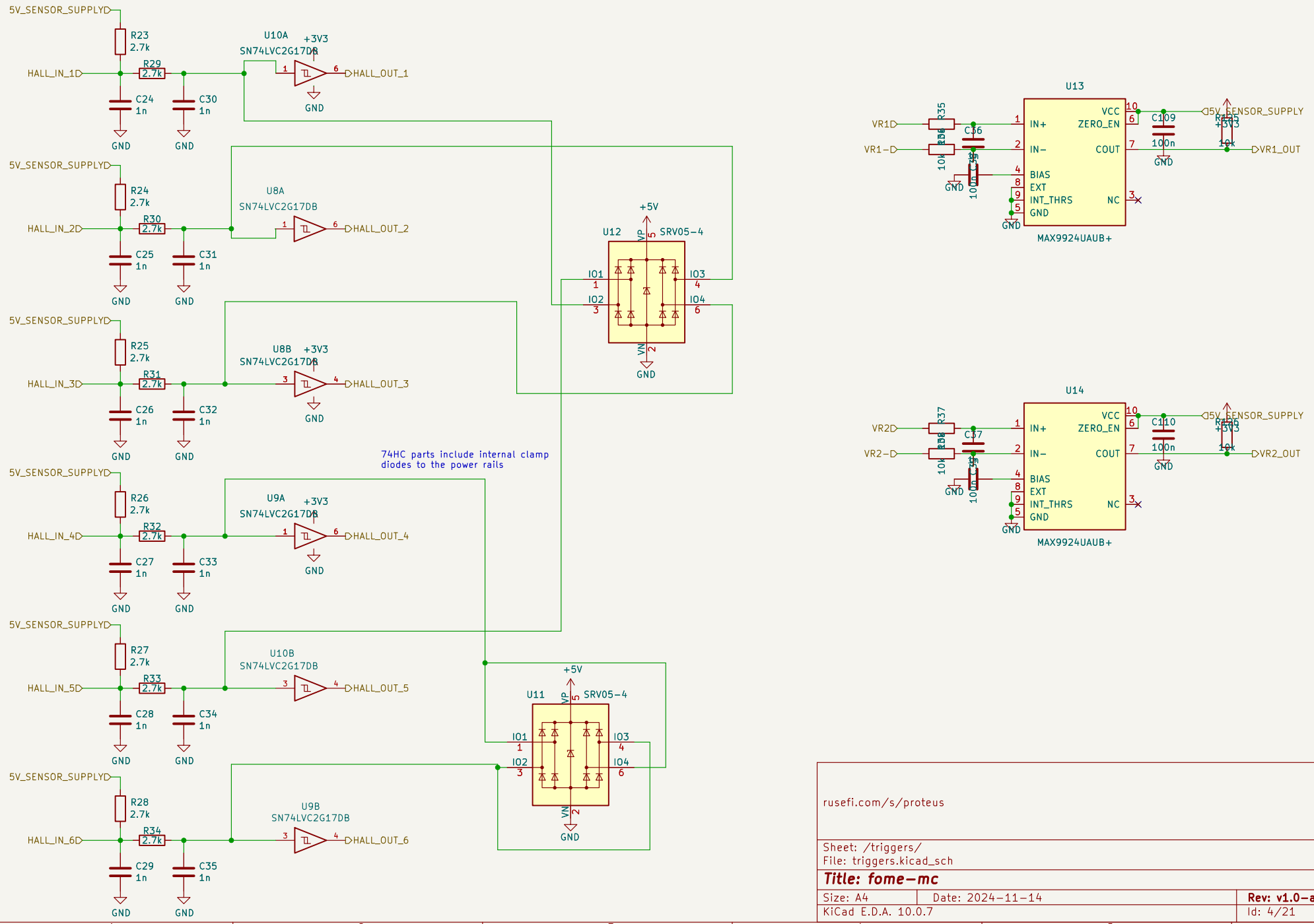


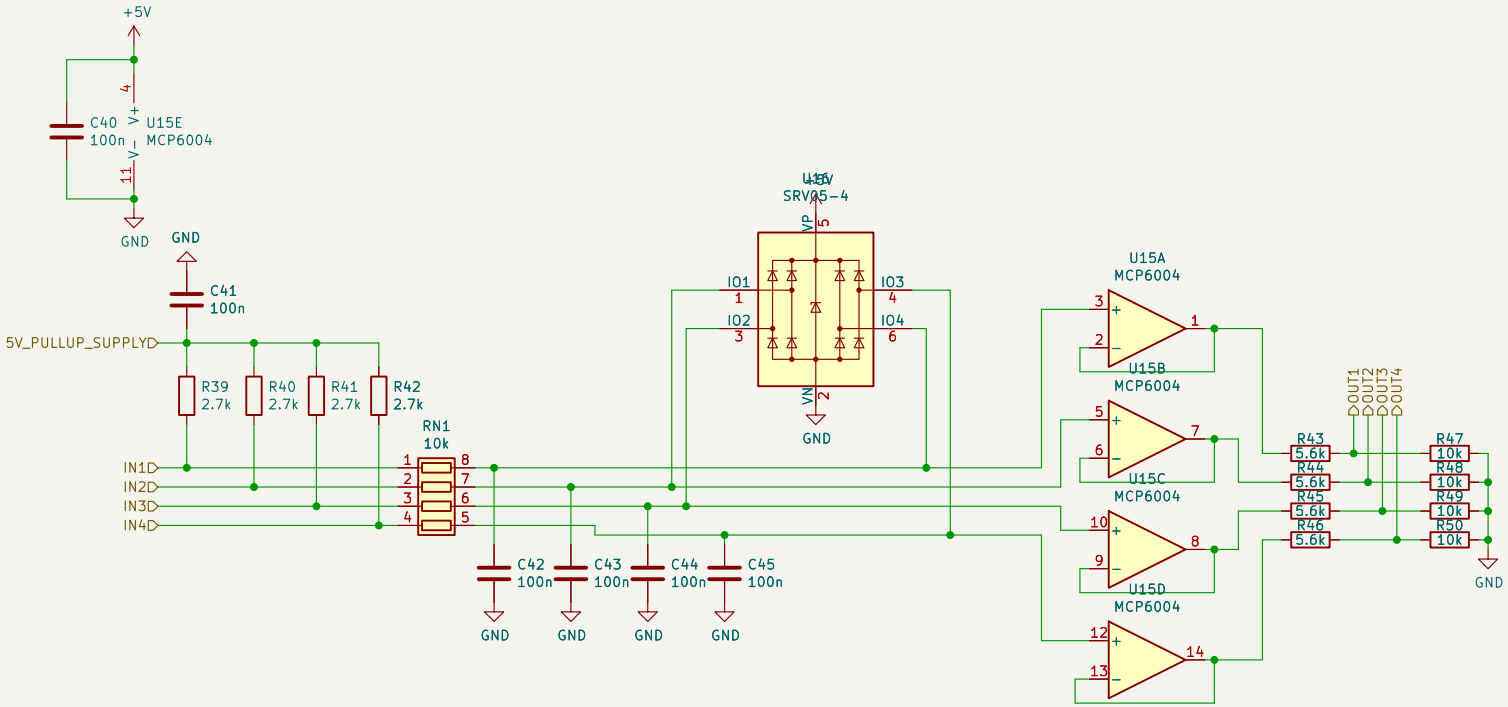
rusefi.com/s/proteus

Sheet: /knock/
File: knock.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 3/21


5V_SENSOR_SUPPLY
HALL_IN_1
HALL_IN_2
HALL_IN_3
HALL_IN_4
HALL_IN_5
HALL_IN_6
R23 2.7k
R29 2.7k
R24 2.7k
R30 2.7k
R25 2.7k
R31 2.7k
R26 2.7k
R32 2.7k
R27 2.7k
R33 2.7k
R28 2.7k
R34 2.7k
C24 1n
C30 1n
C25 1n
C31 1n
C26 1n
C32 1n
C27 1n
C33 1n
C28 1n
C34 1n
C29 1n
C35 1n
GND
U10A
SN74LVC2G17DB
+3V3
HALL_OUT_1
U8A
HALL_OUT_2
U8B
HALL_OUT_3
U9A
HALL_OUT_4
U10B
HALL_OUT_5
U9B
HALL_OUT_6
74HC parts include internal clamp diodes to the power rails
U12
SRV05-4
+5V
IO1
IO2
IO3
IO4
VP
VN
U11
U13
MAX9924UAUB+
VR1
VR2
R35
R36
R37
R38
C36
C37
C38
C39
10k
100n
IN+
IN−
VCC
ZERO_EN
COUT
BIAS
EXT
INT_THRS
NC
C109
C110
R39
R40
10k
VR1_OUT
VR2_OUT
rusefi.com/s/proteus
Sheet: /triggers/
File: triggers.kicad_sch
Title: fome-mc
Size: A4
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 4/21

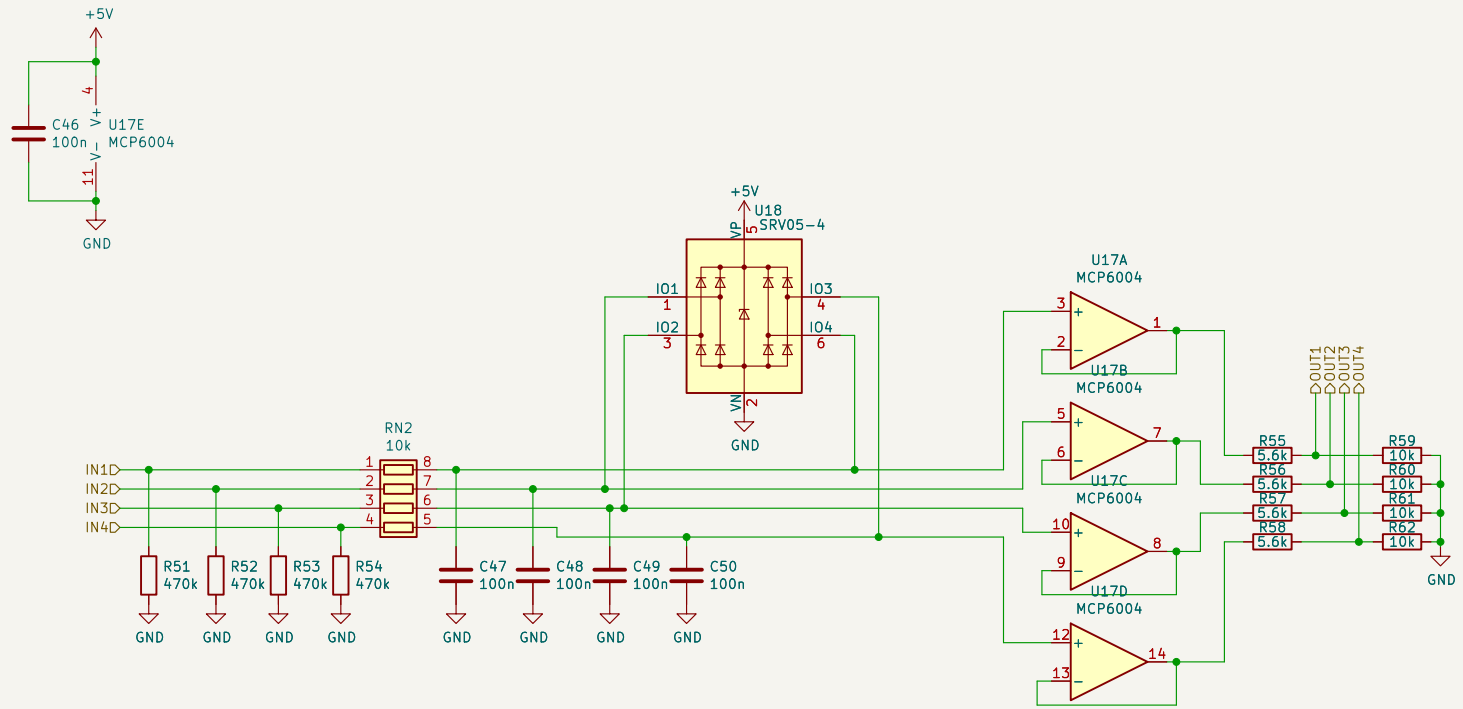


rusefi.com/s/proteus

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 5/21

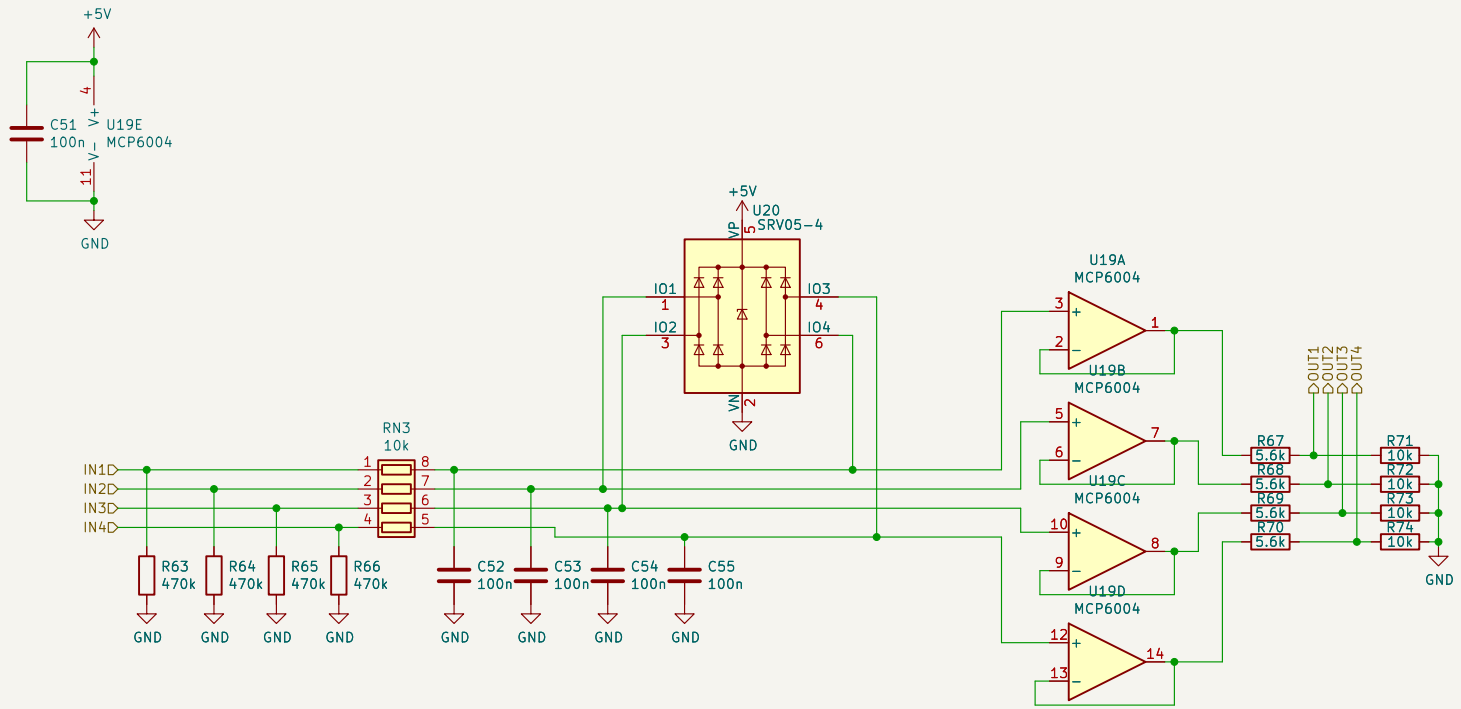


rusefi.com/s/proteus

Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 6/21

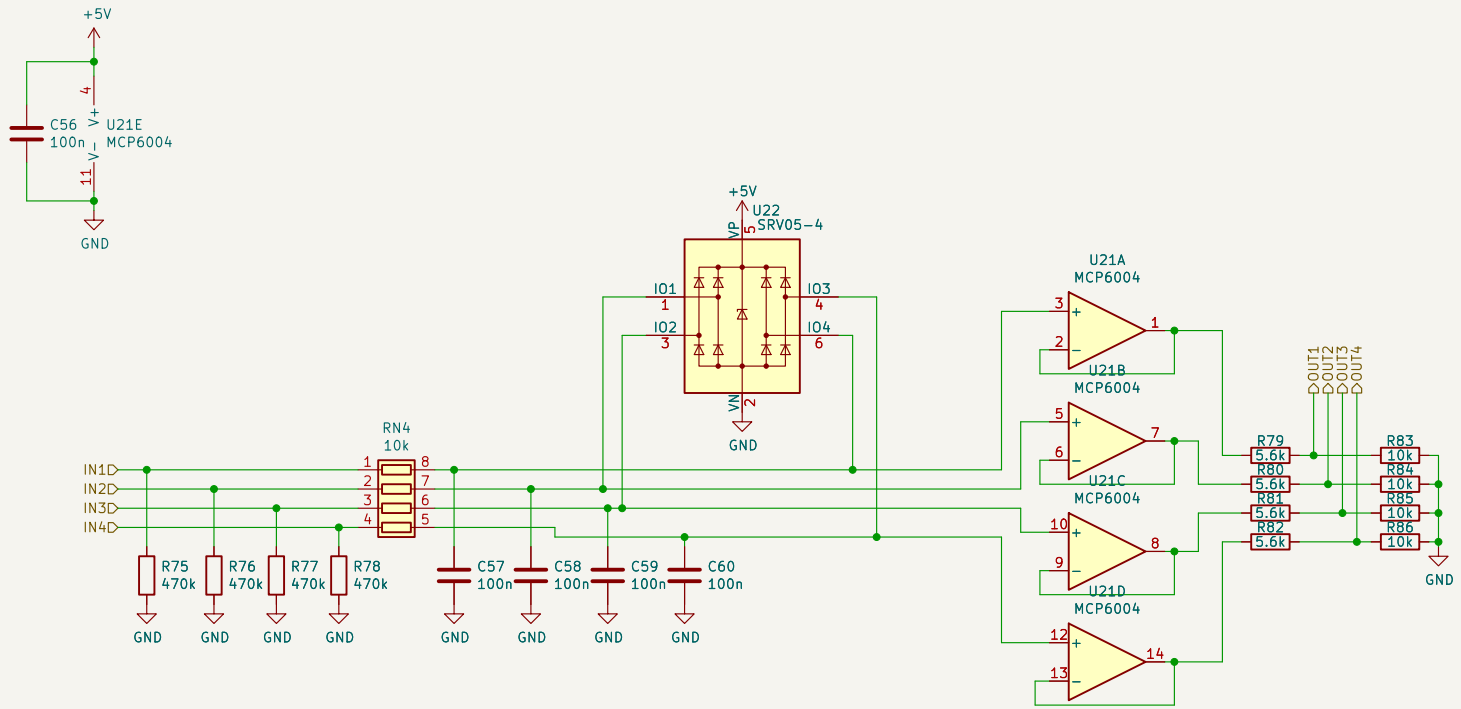


rusefi.com/s/proteus

Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 7/21

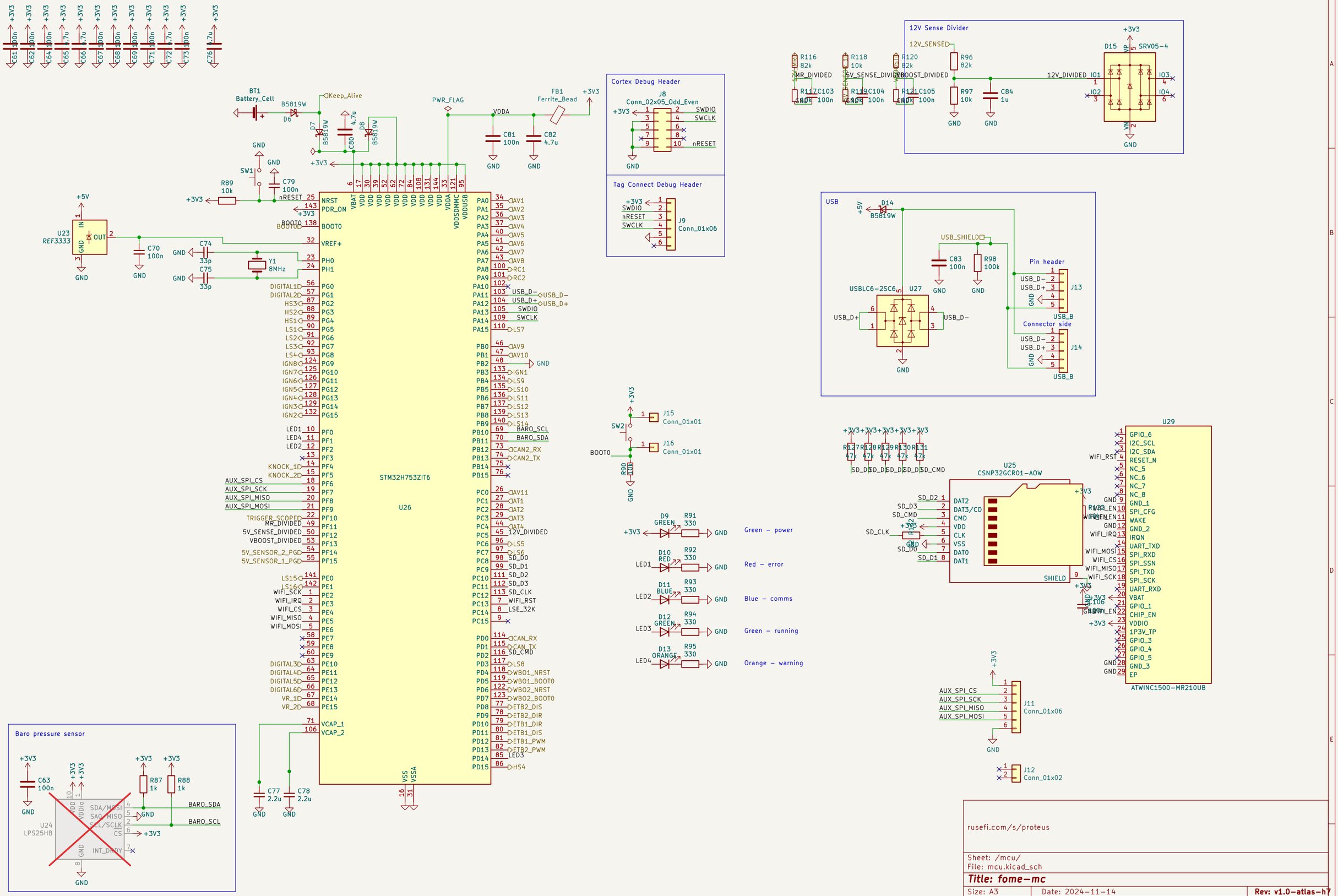


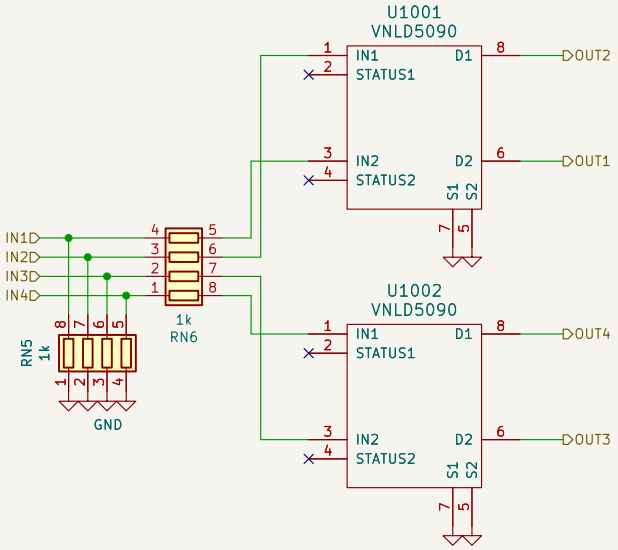
rusefi.com/s/proteus

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 8/21


STM32H753ZIT6
U26
Cortex Debug Header
J8
Conn_02x05_Odd_Even
Tag Connect Debug Header
J9
Conn_01x06
12V Sense Divider
D15
SRV05-4
USB
USBLC6-2SC6
U27
Pin header
Connector side
J13
J14
U29
ATWINC1500-MR210UB
U25
CSNP32GCR01-AOW
J11
Conn_01x06
J12
Conn_01x02
Baro pressure sensor
U24
LPS25HB
Green – power
Red – error
Blue – comms
Green – running
Orange – warning
rusefi.com/s/proteus
Sheet: /mcu/
File: mcu.kicad_sch
Title: fome-mc
Size: A3
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 9/21



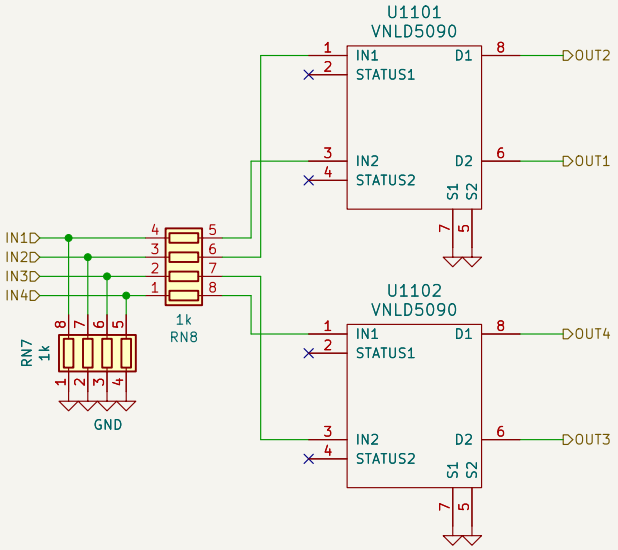
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 10/21



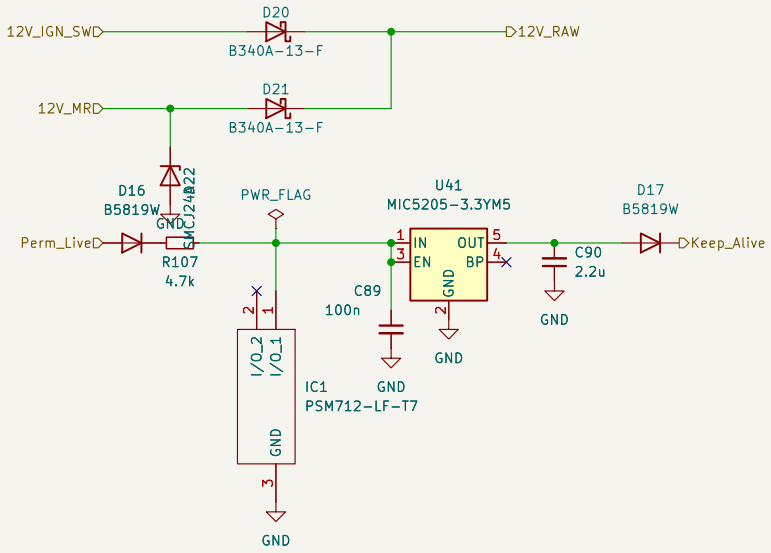
Locate output caps near connector pin

rusefi.com/s/proteus

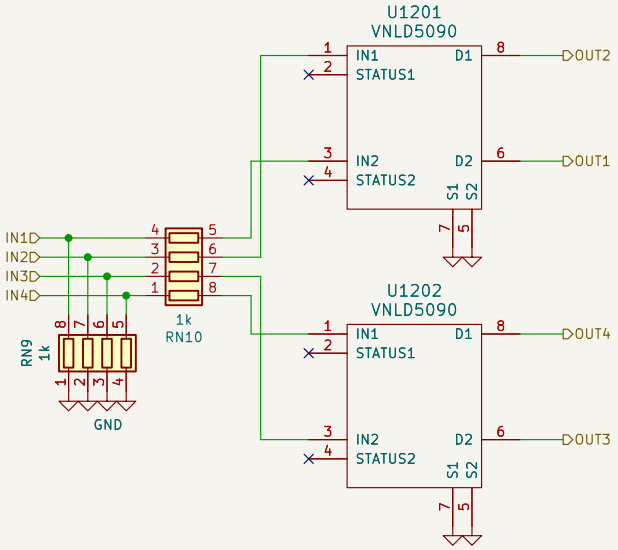
Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 11/21



Sheet: /Keep_Alive/		
File: Keep_Alive.kicad_sch		
Title: fome-mc		
Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 12/21



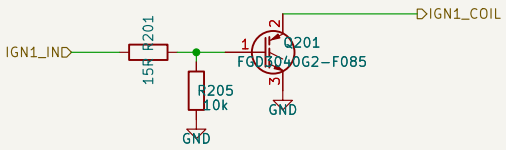
Locate output caps near connector pin

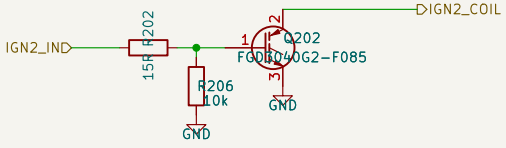
rusefi.com/s/proteus

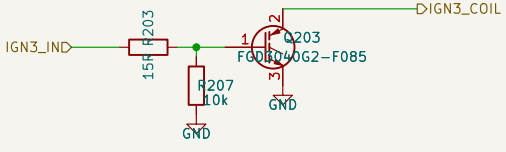
Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

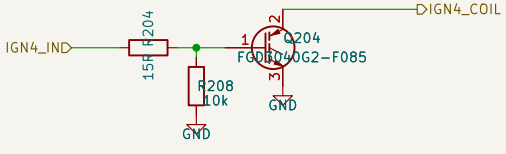
Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 13/21





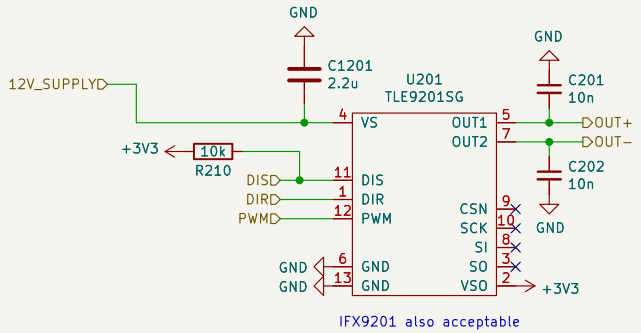




Sheet: /ign_igbt/
File: ign_igbt.kicad_sch

Title: fome-mc 4x High-Current IGBT Ignition Stage

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 14/21

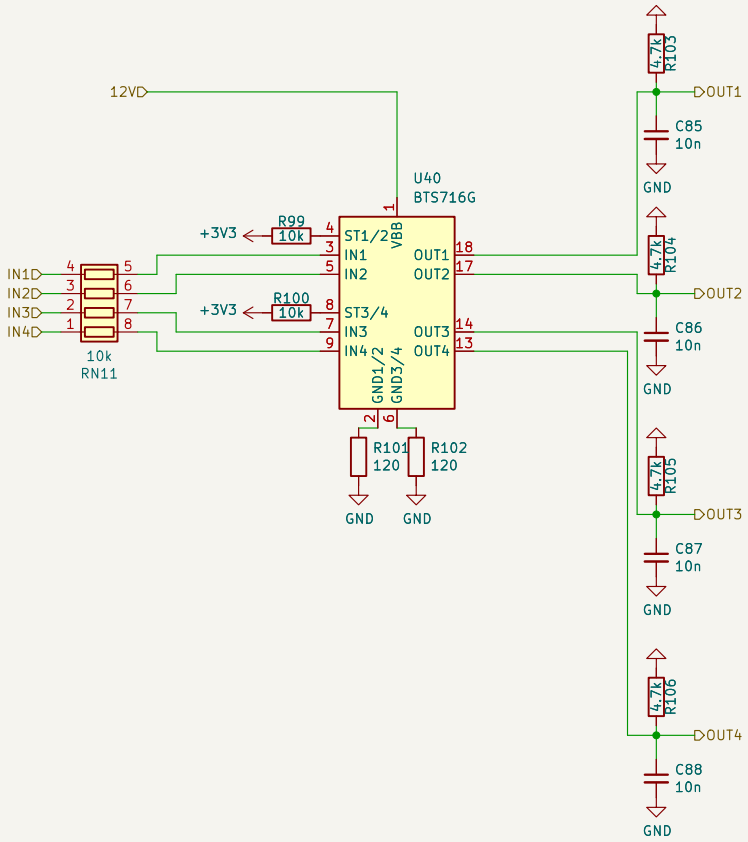


rusefi.com/s/proteus
github.com/mck1117/proteus
rusEFI

Sheet: /etb1/
File: etb-9201.kicad_sch

Title: rusEfi Proteus

Size: A4	Date: 2022-04-09	**Rev: v0.7**
KiCad E.D.A. 10.0.7		Id: 15/21

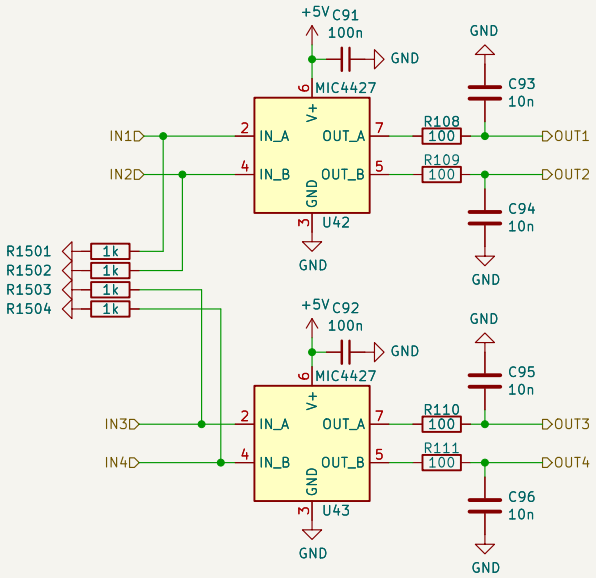


rusefi.com/s/proteus

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 16/21

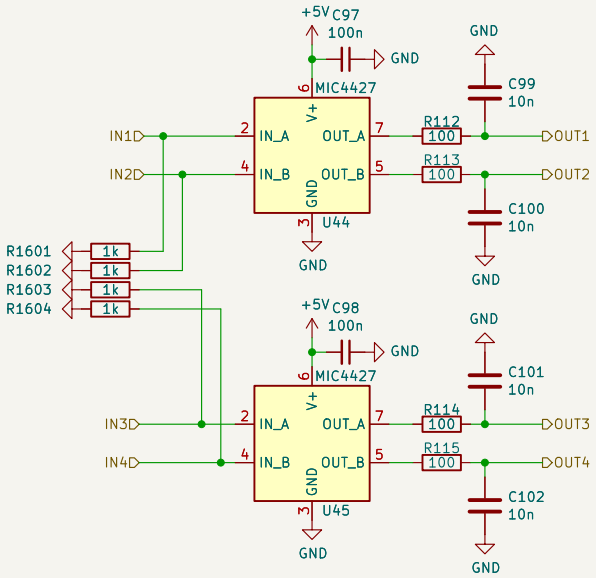


rusefi.com/s/proteus

Sheet: /ign1/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 17/21

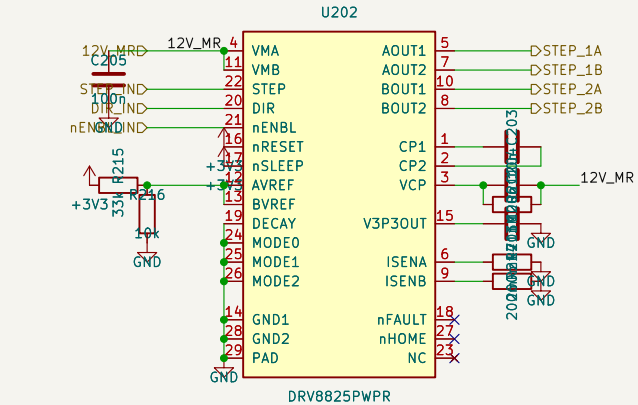


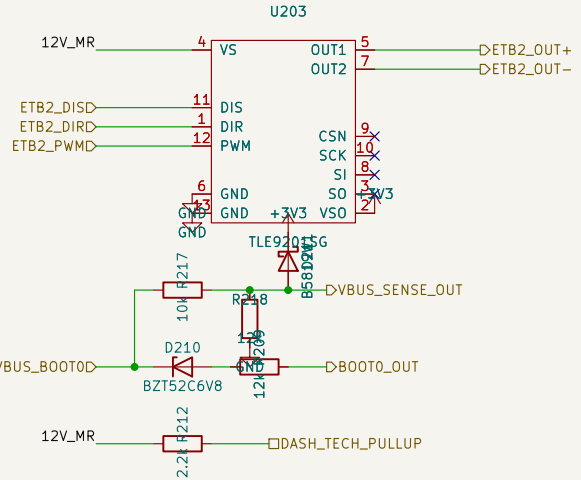
rusefi.com/s/proteus

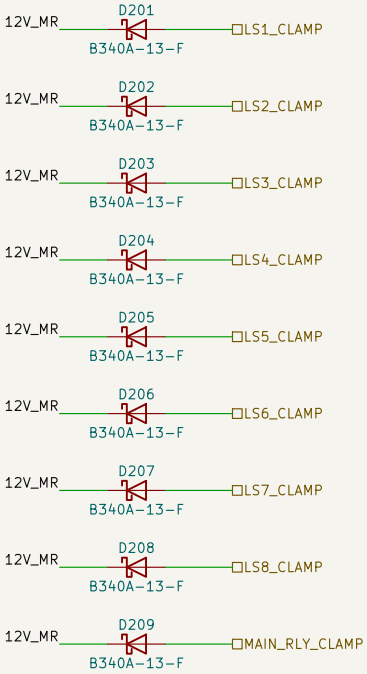
Sheet: /ign2/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 18/21



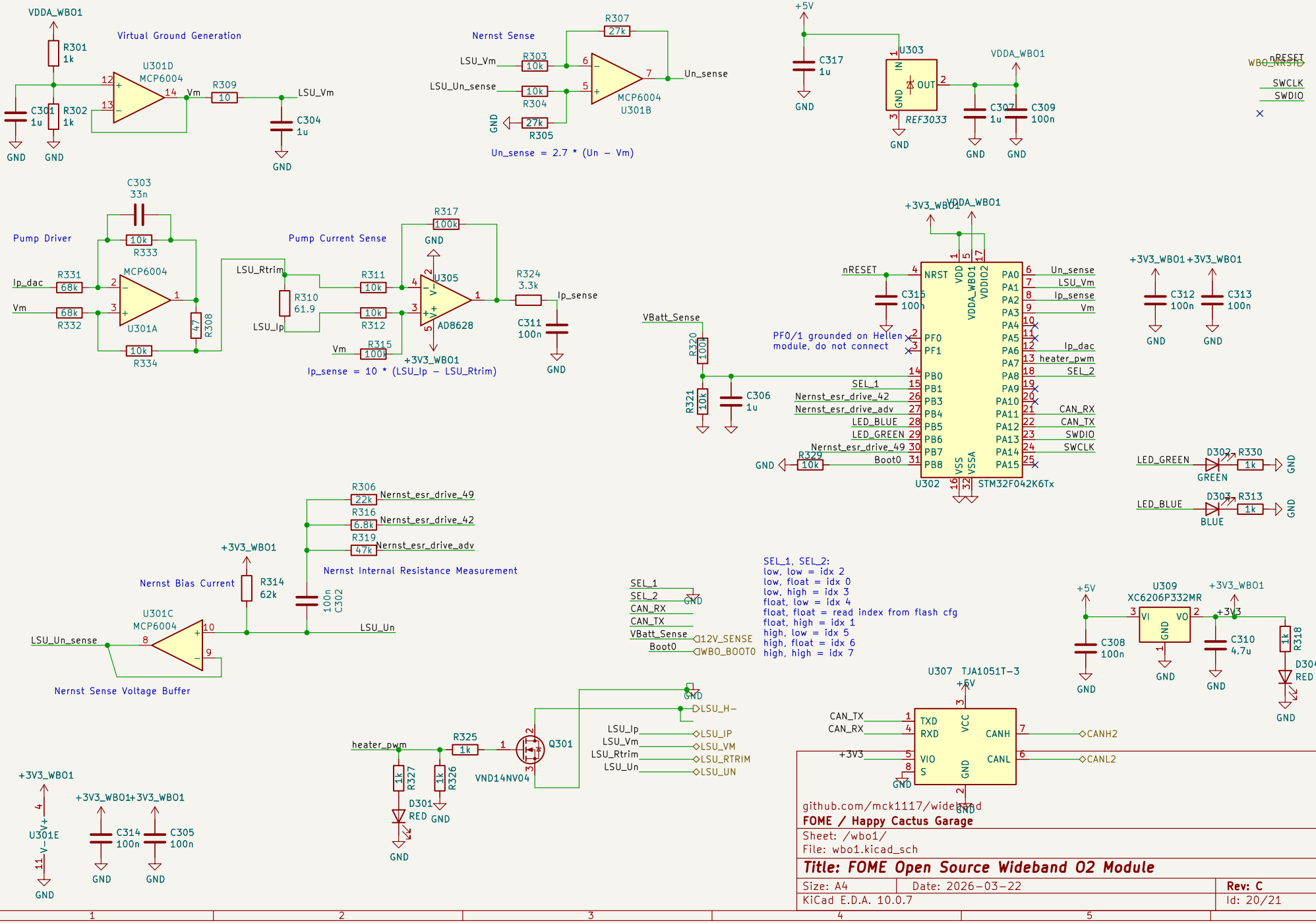


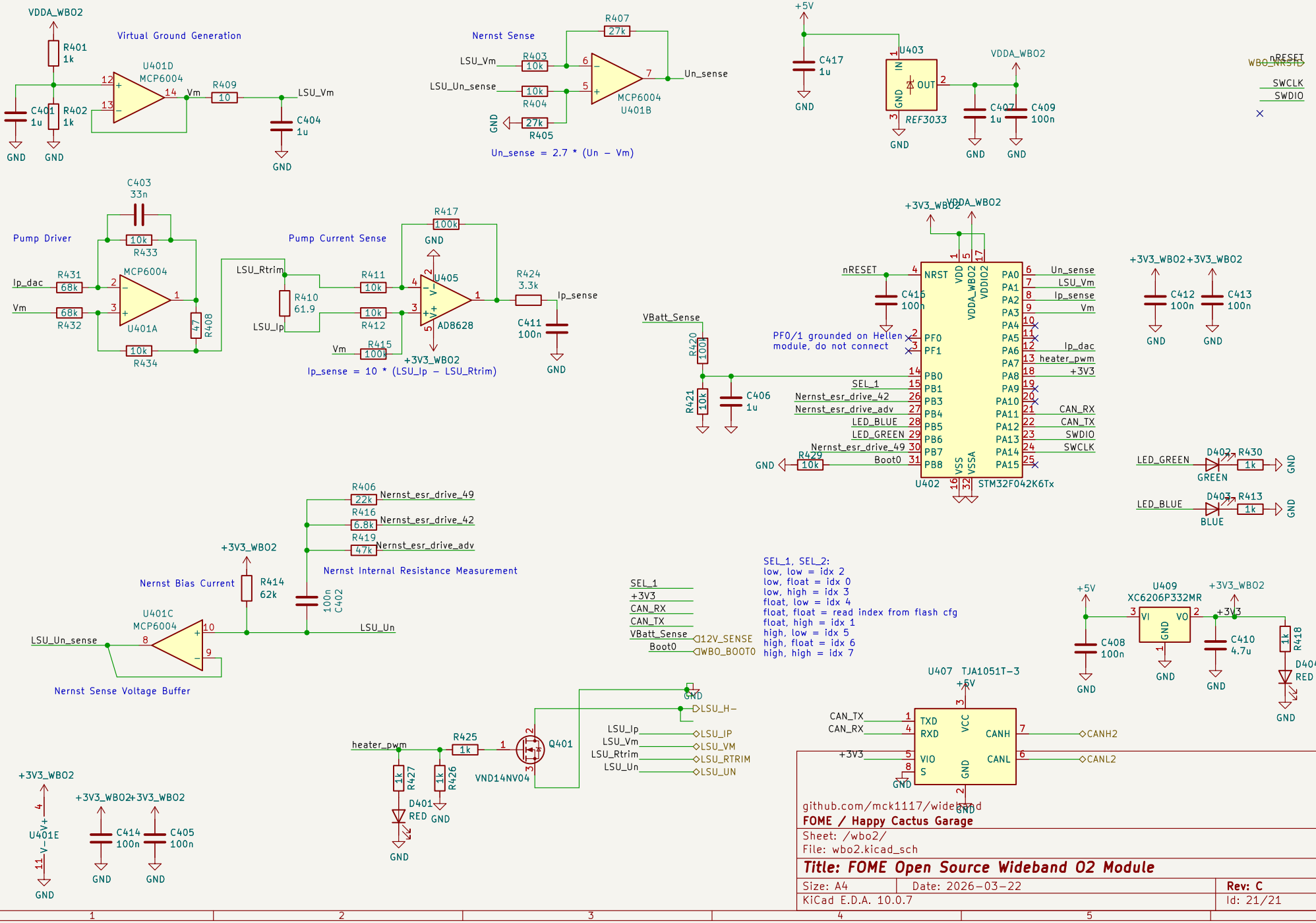


Sheet: /stepper_etb/
File: stepper_etb.kicad_sch

Title: fome-mc Stepper Driver + ETB2 H-Bridge + Low-Side Clamps + BOO

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 19/21


Virtual Ground Generation
VDDA_WBO1
R301 1k
C301 1u
R302 1k
U301D MCP6004
R309 10
C304 1u
LSU_Vm
Vm
GND
Nernst Sense
R307 27k
R303 10k
R304 10k
R305 27k
LSU_Un_sense
Un_sense
U301B MCP6004
Un_sense = 2.7 * (Un − Vm)
+5V
C317 1u
U303 REF3033
C307 1u
C309 100n
nRESET
SWCLK
SWDIO
Pump Driver
C303 33n
R333 10k
R331 68k
R332 68k
R334 10k
R308 47
U301A MCP6004
Ip_dac
Pump Current Sense
R317 100k
R310 61.9
R311 10k
R312 10k
R315 100k
U305 AD8628
R324 3.3k
C311 100n
Ip_sense
LSU_Rtrim
LSU_Ip
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
+3V3_WBO1
VDDA_WBO1
U302 STM32F042K6Tx
nRESET
C316 100n
PF0/1 grounded on Hellen module, do not connect
VBatt_Sense
R320 100k
R321 10k
C306 1u
R329 10k
Boot0
SEL_1
Nernst_esr_drive_42
Nernst_esr_drive_adv
LED_BLUE
LED_GREEN
Nernst_esr_drive_49
C312 100n
C313 100n
D302 GREEN
R330 1k
D303 BLUE
R313 1k
R306 22k
R316 6.8k
R319 47k
Nernst Internal Resistance Measurement
Nernst Bias Current
R314 62k
C302 100n
U301C MCP6004
LSU_Un_sense
LSU_Un
Nernst Sense Voltage Buffer
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
CAN_RX
CAN_TX
12V_SENSE
WBO_BOOT0
U309 XC6206P332MR
+3V3
C308 100n
C310 4.7u
R318 1k
D304 RED
U307 TJA1051T-3
CANH2
CANL2
heater_pwm
R325 1k
Q301 VND14NV04
R327 1k
R326 1k
D301 RED
LSU_H−
LSU_IP
LSU_VM
LSU_RTRIM
LSU_UN
U301E
C314 100n
C305 100n
github.com/mck1117/widebond
FOME / Happy Cactus Garage
Sheet: /wbo1/
File: wbo1.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 20/21


Virtual Ground Generation
VDDA_WBO2
R401 1k
C401 1u
R402 1k
U401D MCP6004
R409 10
Vm
LSU_Vm
C404 1u
GND
Nernst Sense
R407 27k
LSU_Vm
R403 10k
LSU_Un_sense
R404 10k
R405 27k
MCP6004 U401B
Un_sense
Un_sense = 2.7 * (Un − Vm)
+5V
C417 1u
U403
IN
OUT
GND
REF3033
C407 1u
C409 100n
nRESET
WBO_NRST
SWCLK
SWDIO
Pump Driver
C403 33n
10k R433
Ip_dac
R431 68k
Vm
68k R432
MCP6004
U401A
47 R408
10k R434
Pump Current Sense
R417 100k
LSU_Rtrim
R410 61.9
LSU_Ip
R411 10k
R412 10k
R415 100k
U405
AD8628
+3V3_WBO2
R424 3.3k
Ip_sense
C411 100n
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
+3V3_WBO2
VDDA_WBO2
C415 100n
nRESET
PF0/1 grounded on Hellen module, do not connect
VBatt_Sense
R420 100k
R421 10k
C406 1u
SEL_1
Nernst_esr_drive_42
Nernst_esr_drive_adv
LED_BLUE
LED_GREEN
Nernst_esr_drive_49
R429 10k
Boot0
U402 STM32F042K6Tx
Un_sense
LSU_Vm
Ip_sense
Vm
Ip_dac
heater_pwm
+3V3
CAN_RX
CAN_TX
SWDIO
SWCLK
+3V3_WBO2 +3V3_WBO2
C412 100n
C413 100n
LED_GREEN
D402 GREEN
R430 1k
LED_BLUE
D403 BLUE
R413 1k
GND
R406 22k
Nernst_esr_drive_49
R416 6.8k
Nernst_esr_drive_42
R419 47k
Nernst_esr_drive_adv
Nernst Internal Resistance Measurement
+3V3_WBO2
Nernst Bias Current
R414 62k
100n C402
U401C MCP6004
LSU_Un_sense
LSU_Un
Nernst Sense Voltage Buffer
SEL_1
+3V3
CAN_RX
CAN_TX
VBatt_Sense
Boot0
12V_SENSE
WBO_BOOT0
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
+5V
U409 XC6206P332MR
+3V3_WBO2
+3V3
C408 100n
C410 4.7u
R418 1k
D404 RED
GND
heater_pwm
R425 1k
Q401
VND14NV04
R427 1k
R426 1k
D401 RED
LSU_H−
LSU_IP
LSU_VM
LSU_RTRIM
LSU_UN
LSU_Ip
LSU_Vm
LSU_Rtrim
LSU_Un
U407 TJA1051T−3
+5V
CAN_TX
CAN_RX
+3V3
TXD
RXD
VIO
S
VCC
CANH
CANL
CANH2
CANL2
+3V3_WBO2
U401E
+3V3_WBO2 +3V3_WBO2
C414 100n
C405 100n
github.com/mck1117/wideband
FOME / Happy Cactus Garage
Sheet: /wbo2/
File: wbo2.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026−03−22
Rev: C
KiCad E.D.A. 10.0.7
Id: 21/21